Improvements and Extensions to Self-Distilled Agentic Reinforcement Learning (SDAR)

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This document proposes four concrete extensions to SDAR [Lu et al., 2026]. Each section follows the same template: (a) statement, (b) prior-work anchoring, (c) what would test it, (d) prototype pointer. Notation matches the math deep dive (`02-math-deep-dive.md`); in particular, $\Delta_t = \log \pi_T(y_t \mid s_t^+) - \log \pi_\theta(y_t \mid s_t)$ is the teacher–student gap, $g_t = \sigma(\beta\, \mathrm{score}_t)$ is the detached gate, and the SDAR objective is $\mathcal{L}(\theta) = \mathcal{L}_{\mathrm{GRPO}}(\theta) + \lambda_{\mathrm{SDAR}} \cdot \mathcal{L}_{\mathrm{SDAR}}(\theta)$ with $\mathcal{L}_{\mathrm{SDAR}} = \mathrm{Agg}(g_t \Delta_t)$.

1. Mathematical Improvements: a bias bound for gap gating

Statement. Section 7 of the math deep dive sketches that gap-gating reduces bias relative to ungated OPSD, but no closed form is given. We propose the following.

Assumption 1 (Asymmetric reliability of the gap distribution)**.** *Let $p(\Delta)$ denote the marginal distribution of Δ_t over training tokens (under the joint trajectory distribution induced by π_θ and the environment). There exists a threshold $\tau \le 0$ such that the mass $\int_{-\infty}^{\tau} p(\Delta)\,\mathrm{d}\Delta$ is dominated by* skill-retrieval failures *— the privileged context s_t^+ contains text that misled the teacher branch, so $\Delta < \tau$ is unreliable distillation signal. Mass with $\Delta \ge \tau$ corresponds to reliable endorsement (or near-zero disagreement).*

This is the formalisation of paper §1's three failure modes (skill quality, skill utilisation, multi-turn drift), all of which manifest as *negative-tail* mass in $p(\Delta)$.

Proposition 1 (Bias bound for the gated estimator)**.** *Let $\hat{g}_{ungated}(\theta) = \mathbb{E}_{\Delta\sim p}[\Delta]$ be the per-token mean of the ungated SDAR contribution and let $\hat{g}_{gated}(\theta) = \mathbb{E}_{\Delta\sim p}[g(\Delta)\Delta]$ with $g(\Delta) = \sigma(\beta\Delta)$. Define the* ideal *estimator $\hat{g}_{true}(\theta) = \int_{\tau}^{\infty} \Delta\, p(\Delta)\,\mathrm{d}\Delta$ (the contribution restricted to reliable mass). Then*

$$\left|\, \hat{g}_{gated}(\theta) - \hat{g}_{true}(\theta) \,\right| \;\le\; \underbrace{\int_{-\infty}^{\tau} (1-g(\Delta))\,|\Delta|\,p(\Delta)\,\mathrm{d}\Delta}_{\textit{negative-tail leakage}} \;+\; \underbrace{\int_{\tau}^{\infty} (1-g(\Delta))\,\Delta\,p(\Delta)\,\mathrm{d}\Delta}_{\textit{reliable-tail attenuation}}.$$

Both terms vanish as $\beta \to \infty$ (hard step at zero), with rate $O(\exp(-\beta|\tau|))$ for the first and $O(\beta^{-1})$ for the second.

Prior work. The decomposition mirrors the baseline-doesn't-bias proof of GRPO [Shao et al., 2024] and the chain Ch. 31 derivation, where a state-dependent baseline is shown to be zero-mean against the policy gradient. The gate g_t is detached, so the same conditional-expectation argument applies: gating shifts the *target* but does not introduce a new dependence on θ. The asymmetric-tail framing is inspired by importance-sampling truncation analyses common in PPO [Schulman et al., 2017] and InstructGPT-style RLHF [Ouyang et al., 2022].

What would test it. The full statement and proof appear in `proofs/gating-bias-bound.tex`. Empirically, the toy bandit in `sandbox/toy_sdar.py` (and the comparison in `improvements/adaptive-gate.py`) lets us measure the empirical bias of the gated vs. ungated estimator under a synthetic $p(\Delta)$ with controllable negative-tail mass.

Prototype pointer. PROOF: `proofs/gating-bias-bound.tex` (stub). Numerical sanity check: `improvements/adaptive-gate.py` already computes per-token gate-fire rates; extending to bias estimates is ∼10 lines.

2. Code / Implementation Improvements: adaptive-quantile gating

Statement. The paper uses a fixed sigmoid $g_t = \sigma(\beta\,\Delta_t)$ centred at $\Delta_t = 0$. We propose

$$g_t \;=\; \sigma\big(\beta\,(\Delta_t - \hat{q}_\tau)\big), \qquad \hat{q}_\tau \leftarrow (1-\eta)\,\hat{q}_\tau + \eta\,\Delta_t^{\text{thresh}},$$

where $\hat{q}_\tau$ is an exponentially-moving estimate of the τ-quantile (e.g. $\tau = 0.3$) of recent gaps. Rationale: the gap distribution drifts over training as the student policy improves; the *absolute* criterion $\Delta_t > 0$ stops being meaningful after a few hundred steps because typical gaps shift toward zero. A relative criterion "fire on the top $1-\tau$ fraction" stays well-calibrated.

Prior work. Quantile-based truncation has a long history in robust statistics; the closest deep-RL precedent is V-Trace-style importance-clipping in IMPALA, and the running-quantile baselines in Wang et al. [2026]. The detached-gate structure is preserved, so the convergence properties argued in Prop. 1 still apply.

What would test it. Three-way comparison on the toy MDP: GRPO baseline, fixed-threshold SDAR ($\sigma(2\Delta_t)$), and adaptive-quantile SDAR. Metric: final reward subject to a per-turn KL budget identical to the paper's stability criterion.

Prototype pointer. MEASUREMENT: `improvements/adaptive-gate.py`. Returns `{fixed_threshold_rewa` `adaptive_kl, gate_fire_rate_fixed, gate_fire_rate_adaptive}`.

3. Experimental Extensions: λ_{SDAR} sensitivity sweep

Statement. The SDAR ablation in the paper fixes λ_{SDAR} at one value and varies the gating strategy. But λ_{SDAR} is itself a critical hyperparameter: $\lambda_{\text{SDAR}} = 0$ recovers GRPO and gives no distillation; $\lambda_{\text{SDAR}} \to \infty$ overwhelms the RL signal and the policy collapses to a maximum-likelihood estimate of the teacher. Somewhere between is an optimum. We propose a 5-point sweep $\lambda_{\text{SDAR}} \in \{0, 0.1, 0.3, 1, 3\}$ at fixed β, reporting both final reward and final per-turn KL, and identifying the constrained optimum

$$\lambda^\star \;=\; \arg\max_{\lambda} \text{reward}(\lambda) \quad \text{s.t. } \text{kl}(\lambda) \le 2 \cdot \text{kl}(0).$$

Prior work. Standard auxiliary-loss-weight sweeps appear in DeepSeek-AI [2025] and Zhao et al. [2026]; both observe an inverted-U in reward as the auxiliary weight grows. Our sweep is the agentic-RL specialisation of this exercise.

What would test it. Five 100-step training runs on the toy MDP, one per λ value, with $\beta = 2$ held constant. Constrained $\arg\max$ over the resulting (reward, KL) Pareto trace.

Prototype pointer. MEASUREMENT: `improvements/lambda-sweep.py`. Returns the full `{sweep, optimal_lambda}` dictionary.

4. Theoretical / Conceptual Connections: SDAR as constrained RL

Statement. Frame the SDAR objective as a *constrained* GRPO problem with a per-token trust constraint:

$$\max_{\theta}\; J_{\text{GRPO}}(\theta) \quad \text{s.t. } \; \mathbb{E}_t\big[\,\mathbf{1}[t \in \mathcal{T}]\; D_{\text{KL}}(\pi_\theta(\cdot \mid s_t)\,\|\,\pi_T(\cdot \mid s_t^+))\big] \;\le\; \delta,$$

where $\mathcal{T}$ is the set of *trusted* tokens (those for which the privileged context endorses the student trajectory). The Lagrangian is

$$\mathcal{L}(\theta, \mu) \;=\; J_{\text{GRPO}}(\theta) \;-\; \mu \cdot \Big(\mathbb{E}_t[\mathbf{1}[t \in \mathcal{T}] \cdot D_{\text{KL}}^{(t)}] - \delta\Big).$$

SDAR's gated loss has *exactly* this structure with the global multiplier μ replaced by a state-dependent per-token multiplier $\lambda_{\text{SDAR}} \cdot g_t$. In other words, SDAR is a *soft, primal-only* approximation to a per-token trust-constrained RL problem, where the gate g_t is an indicator surrogate for membership in $\mathcal{T}$ and λ_{SDAR} is a fixed multiplier that plays the role of the dual variable.

Prior work. Constrained RL with primal-dual saddle-point solvers traces back to the option/subgoal formalism of **?**. The chain Ch. 30 max-entropy-RL derivation gives the same Lagrangian form for the entropy-regularised case. The modern primal-only variant (treating μ as a hyperparameter rather than a learned dual variable) is the same shortcut taken by PPO's KL-penalty schedule [Schulman et al., 2017].

What would test it. This is a translation, not a new theorem. The deliverable is a short proof sketch in `proofs/sdar-as-constrained-rl.tex` that (i) writes the SDAR gradient as the gradient of the Lagrangian above with $\mu = \lambda_{\text{SDAR}}$ and the indicator replaced by g_t, and (ii) cites the saddle-point convergence conditions from constrained-RL literature for which SDAR's fixed multiplier is sound (essentially: when the constraint is slack at the optimum, primal-only with a sufficiently large μ works).

Prototype pointer. PROOF: `proofs/sdar-as-constrained-rl.tex` (stub).

Discussion

The four improvements compose. The mathematical bias bound (§1) is the *justification* for adaptive-quantile gating (§2): the bound's negative-tail leakage term shrinks faster when the gate centre tracks the empirical τ-quantile rather than sitting at zero. The λ_{SDAR} sweep (§3), once run jointly with the adaptive gate, would reveal a *joint* optimum $(\lambda^\star, \tau^\star)$, plausibly different from the single-variable optimum. And the constrained-RL framing (§4) gives the λ_{SDAR} sweep its theoretical reading: the optimal $\lambda^\star$ is an estimate of the dual variable that would solve the constrained problem at the desired KL budget. Together they upgrade SDAR from a heuristic auxiliary loss to a principled primal approximation of a per-token trust-constrained agentic-RL problem.

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